

Blockchain Introduction

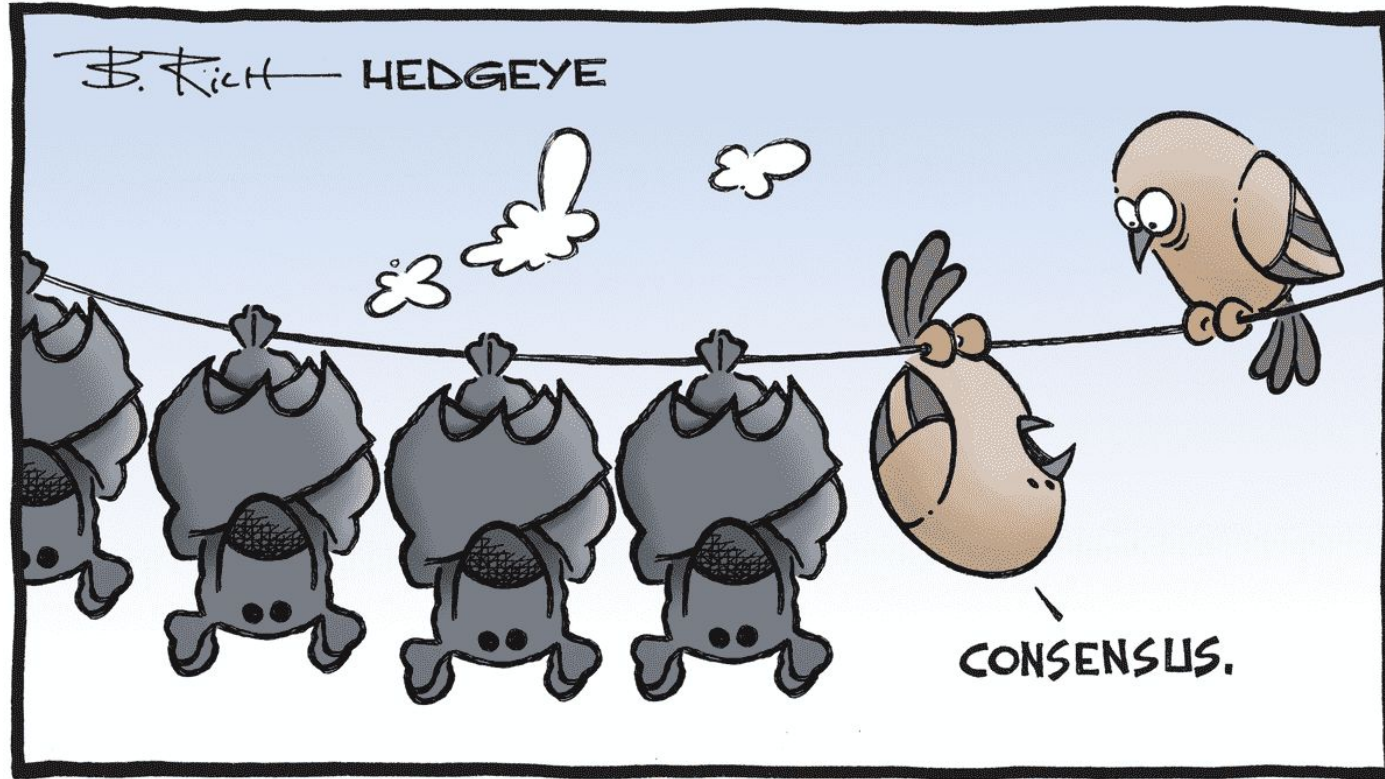
Andrea Vitaletti
vitaletti@diag.uniroma1.it

- A2 15:00-17:00 Thursday
- A7 10:00-13:00 Friday
- In continuous evolution
- Based on what I do consider well done job ... no need to reinvent the wheel
- Comments, suggestions, thoughts, feedback, missing references are very welcome
- <https://classroom.google.com/c/MjMwODQyMTcwODBa?cjc=wkqxriz>
- <https://docs.google.com/spreadsheets/d/133Qh3MWUzqmJIC8TR29X4IZk3NVw0Y4TRHuOa-x55o/edit?gid=2060682927#gid=2060682927>
- <https://andreavitaletti.github.io/BC/>



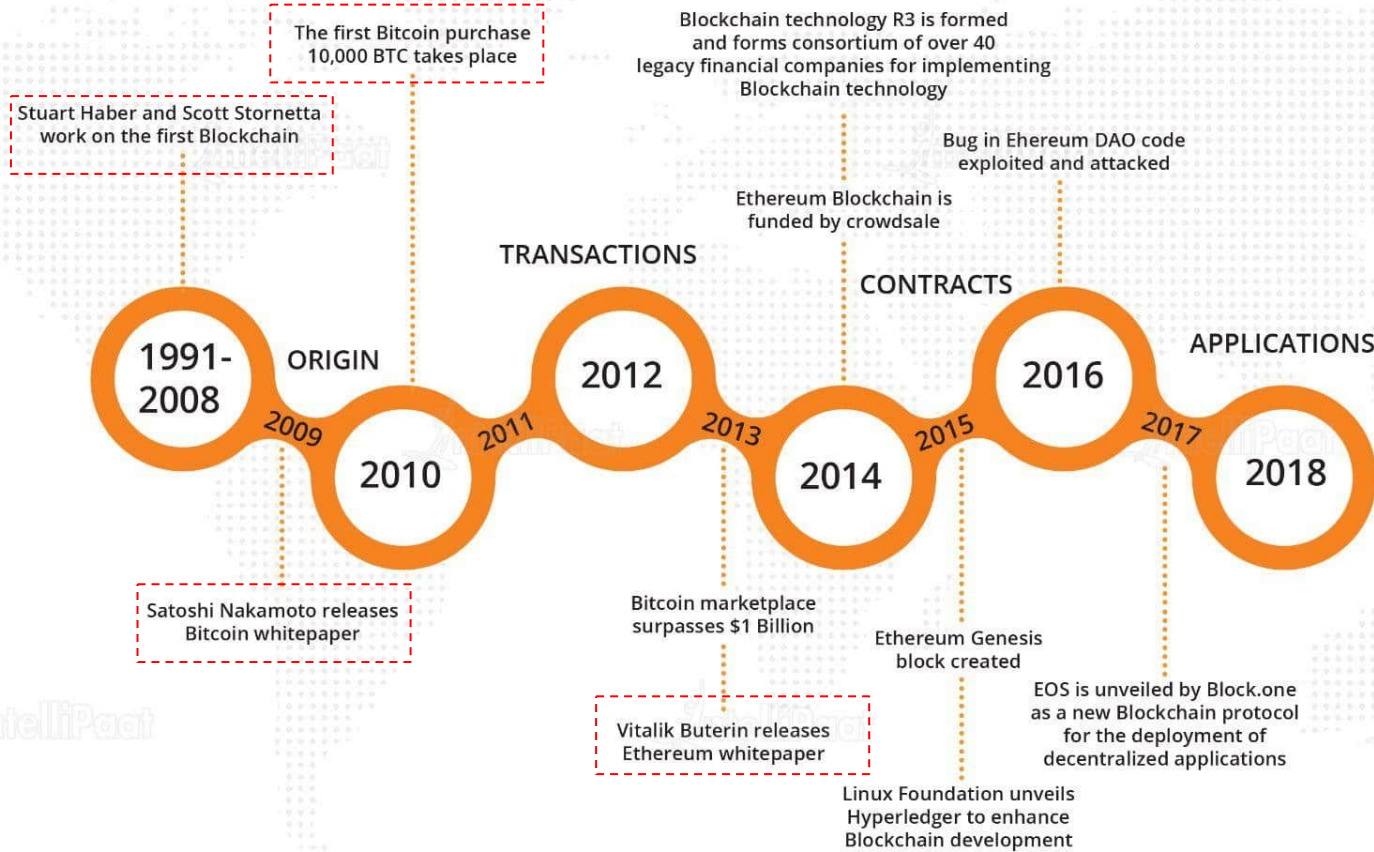
GOAL

Achieve **consensus** on what is a blockchain, the main tech ingredients to implement it and describe its evolution ;-)



https://d1vhlis6iwh515.cloudfront.net/charts/resized/76392/large/04.12.2019_consensus_cartoon.png

THE HISTORY OF BLOCKCHAIN TECHNOLOGY



A transaction: the exchange (or agreement for an exchange) of an asset

TX

EUR

Bitcoin Transaction

Broadcasted on 06 Feb 2025 08:37:11 GMT+1

Hash ID
68e9a945414bc9b7a768be36ea1df992e08ffee5527a30c2b70d4474bfa8f9ec

Amount 0.00180720 BTC • €171.19
Fee 232 SATS • €0.22

From bc1py-6e8nn
To 2 Outputs

Confirmed

 This transaction has 3 Confirmations. It was mined in Block 882,524

 This transaction is efficient, no issues detected.



Financial Assets: the evolution of Money



7th Century
B.C.
Metal Coins



1928
Plastic
Cards



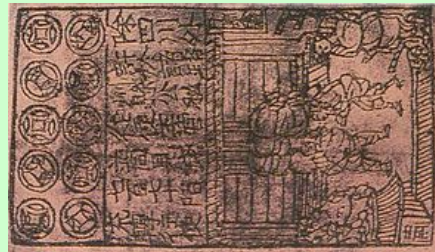
2009
Crypto
Currency

5000 B.C.
Barter
System

4000 B.C.
Gold

812 Paper
Money

1994
Electronic
Money



THE FUTURE OF MONEY



Black Box Bank



Trust “individuals” ... or servers ;-)



**BLACK
BOX**

**“humans in the
loop”**

Consensus from the “society” ... or a distributed protocol!



**WHITE
BOX**

“algorithms”

In the networking world

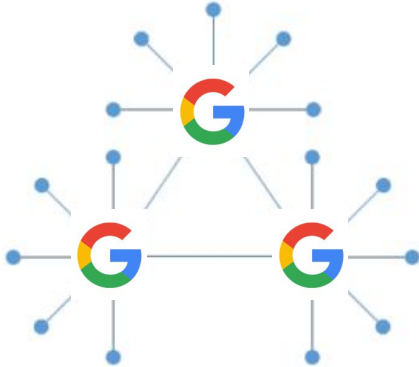
Paul Baran



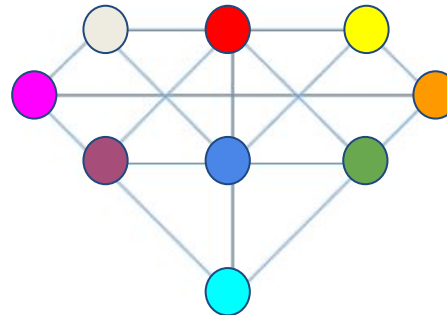
CENTRALIZED



DECENTRALIZED



DISTRIBUTED



source: <https://nvlpubs.nist.gov/nistpubs/jr/2018/NIST.IR.8202.pdf>
thanks to <https://www.youtube.com/watch?v=CEatIMM0QrQ>

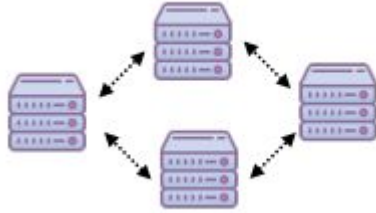


Centralized
Trust



Decentralized
Consensus

Creation



A distributed cryptographic protocol
Money is “minted” (created) according to protocol rules

Authenticity:

- digital signature

CryptoCurrency

Currency



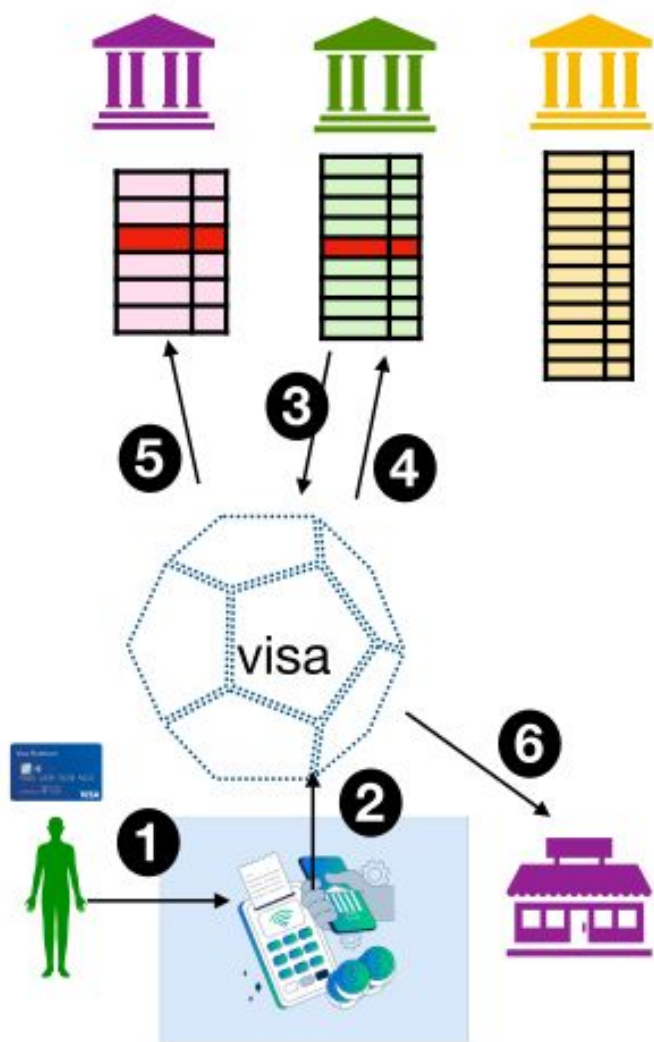
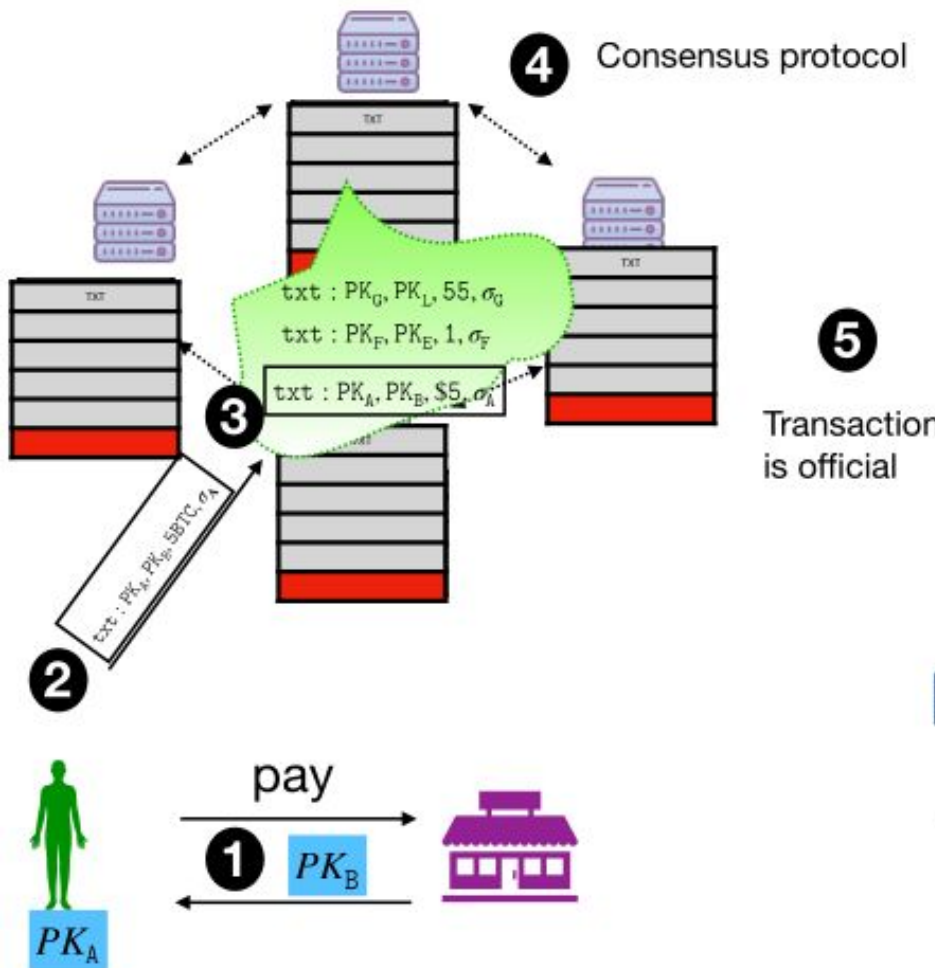
Authenticity:

- bills: Physical watermarking
- digital: bank security systems

Payments

Cryptocurrency

Currency



Correctness

Cryptocurrency

Currency

The protocol is open

We have formal proofs

Everyone can check every transaction on the table

We trust the correctness of banks' systems and gateways

Auditing

Account: public key PKa

Payment: a signature on the amount PKa wants to send to PKb

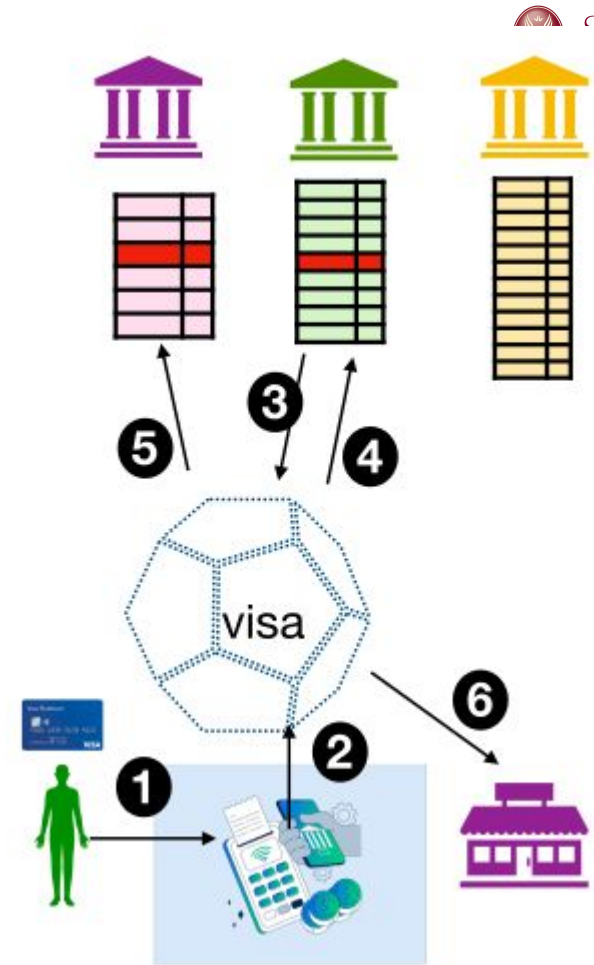
- UTXO and Account model

Correctness: a payment is valid after it is added to the public ledger through a consensus protocol run by all nodes in the network

Privacy: Digital Currency

Who sees the payments:

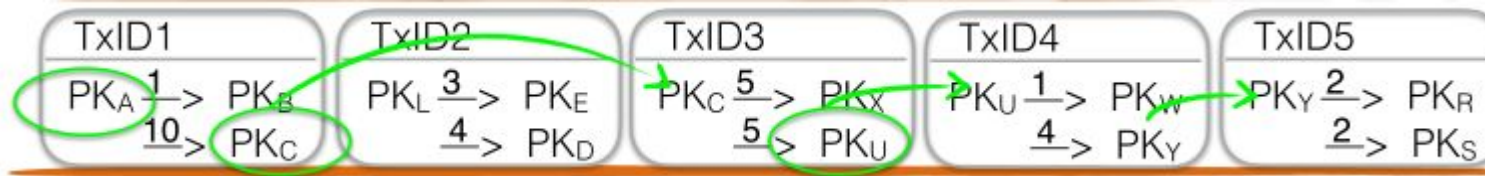
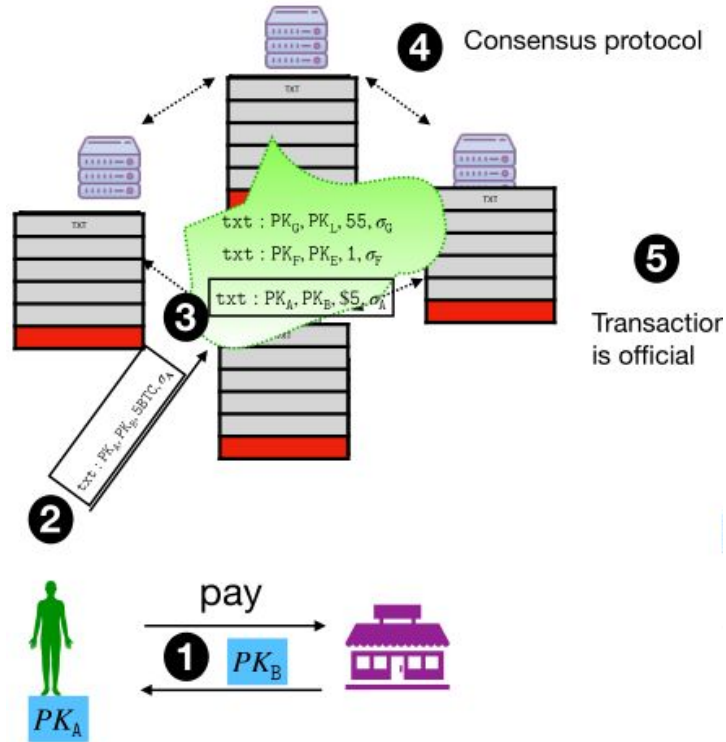
- The source bank
- The destination bank
- The gateways (e.g Visa)

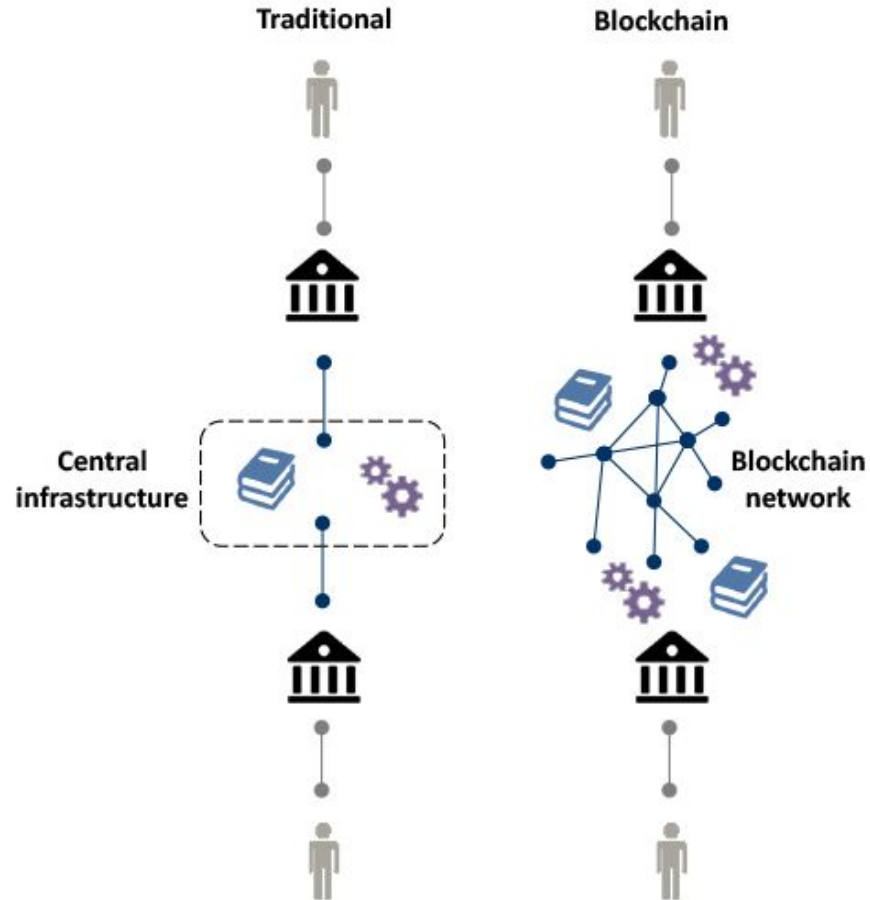


Privacy: Cryptocurrencies

Who sees the payments:

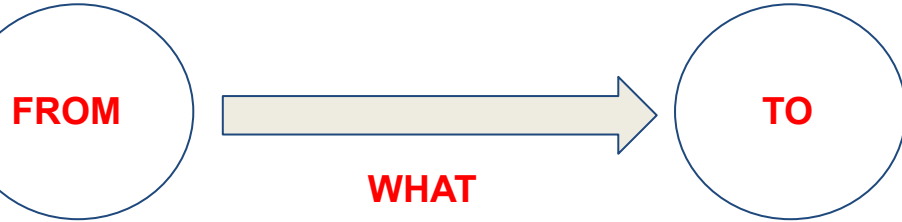
- All



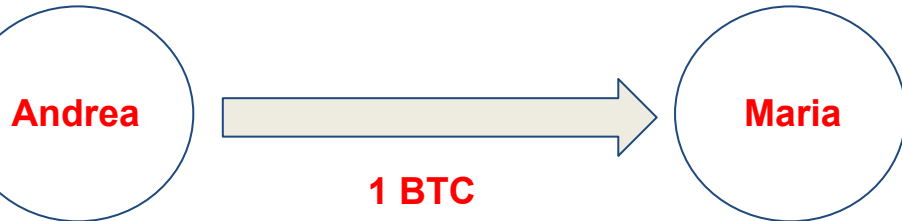


	Purpose	In Traditional networks	In Blockchain networks
Front-end	Interaction with end-user (untouched by blockchain technology)	Not affected: remains the same	
Messaging	Technical connectivity with the network	Through central infrastructure	Peer-to-peer
Processing	Execution of transactions	- Centrally - Batch or per trx	- Decentral - In 'blocks'
Ledger	Keeps track of participants' balances	- Central - Closed (one trusted party)	- Decentral - Public

WHEN

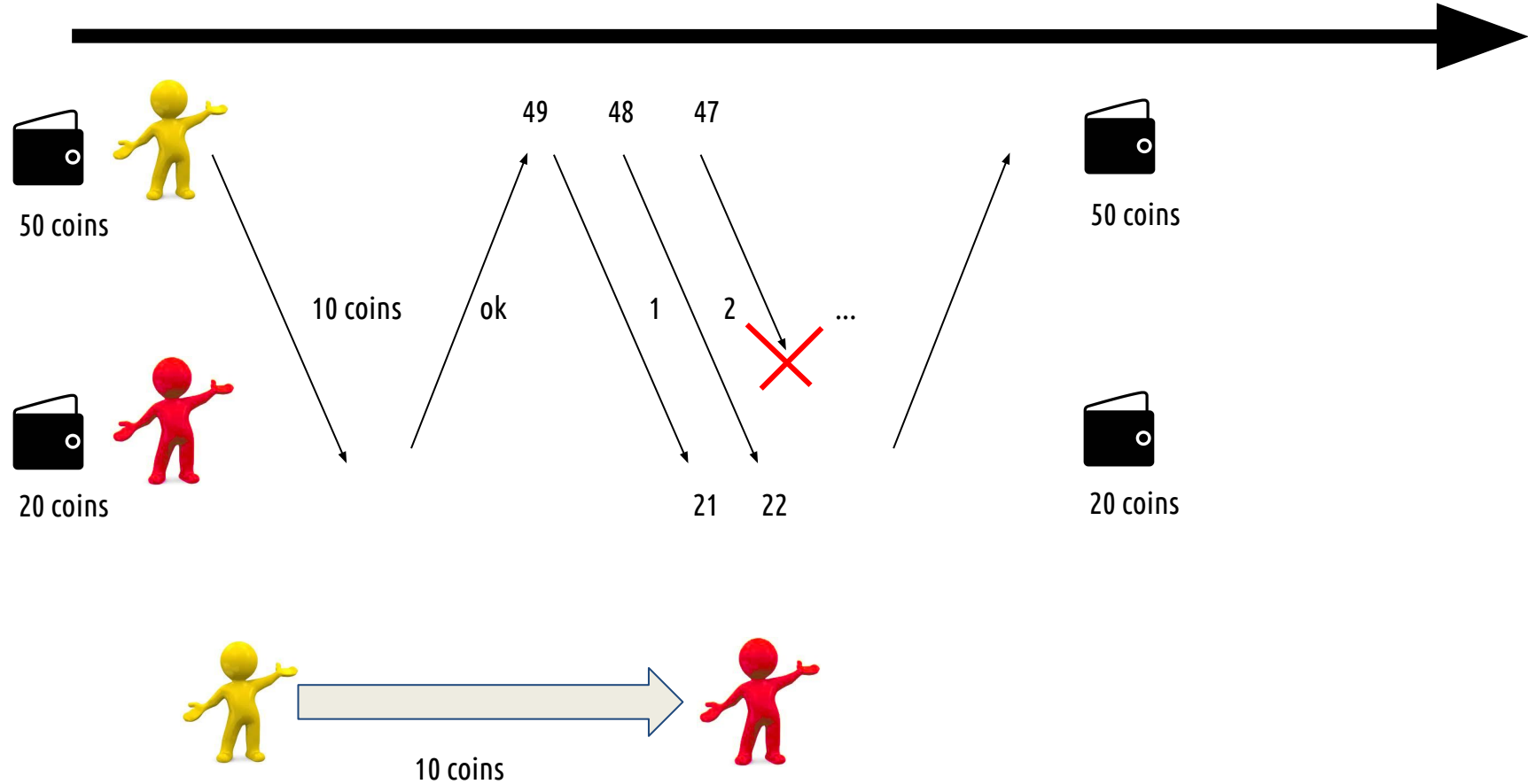


Block



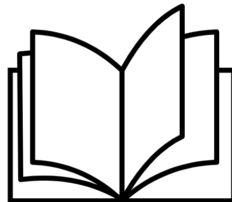
An atomic event supported by the protocol

- Andrea → Maria 1 BTC
- State 1 → State 2 in the world computer



Ledger of transactions

FROM	TO	WHEN	WHAT
A	B	a/b/c	10
B	C	e/f/g	5
B	D	e/f/g	7
...	...	...	...



Ledger of transactions: WHERE

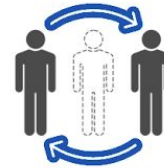
FROM	TO	WHEN	WHAT
A	B	a/b/c	10
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B	D	e/f/g	7
...	...	...	...



intermediaries



trust



disintermediation

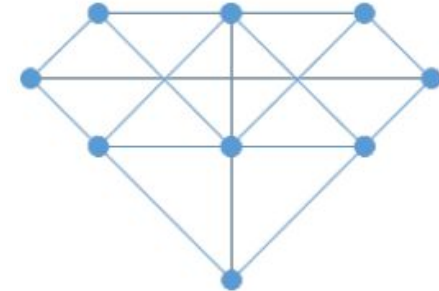


consensus

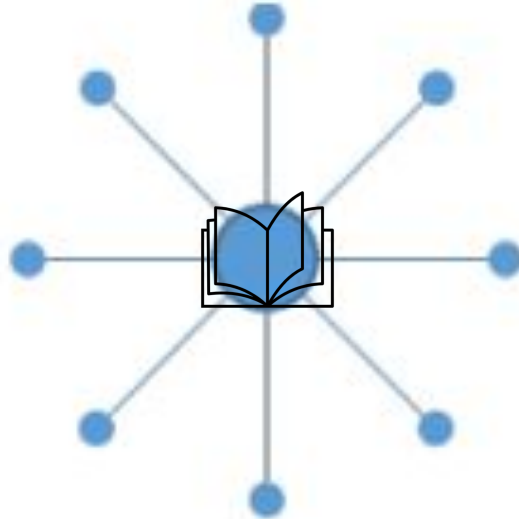
EASY



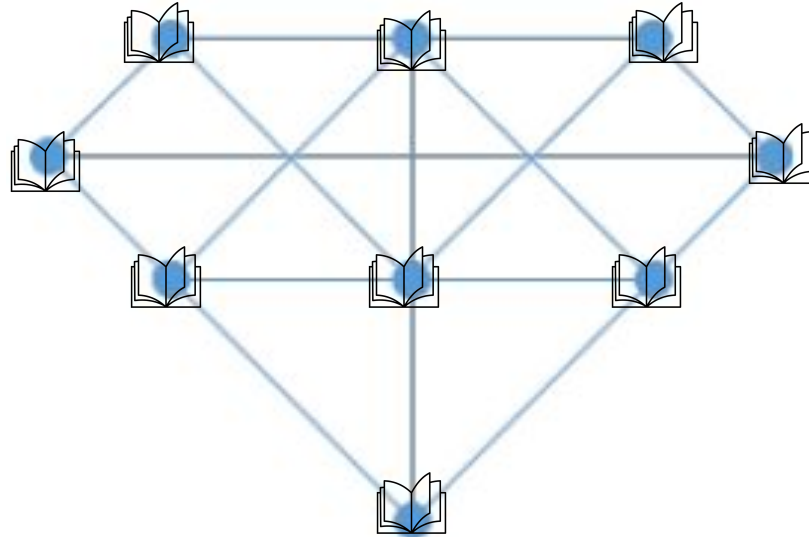
DIFFICULT



**WHERE
CENTRALIZED
EASY**



WHERE DISTRIBUTED



What is blockchain?

DIFFICULT: from trust to consensus (i.e. decentralized governance)

A distributed ledger that can record transactions in an immutable, verifiable and permanent way

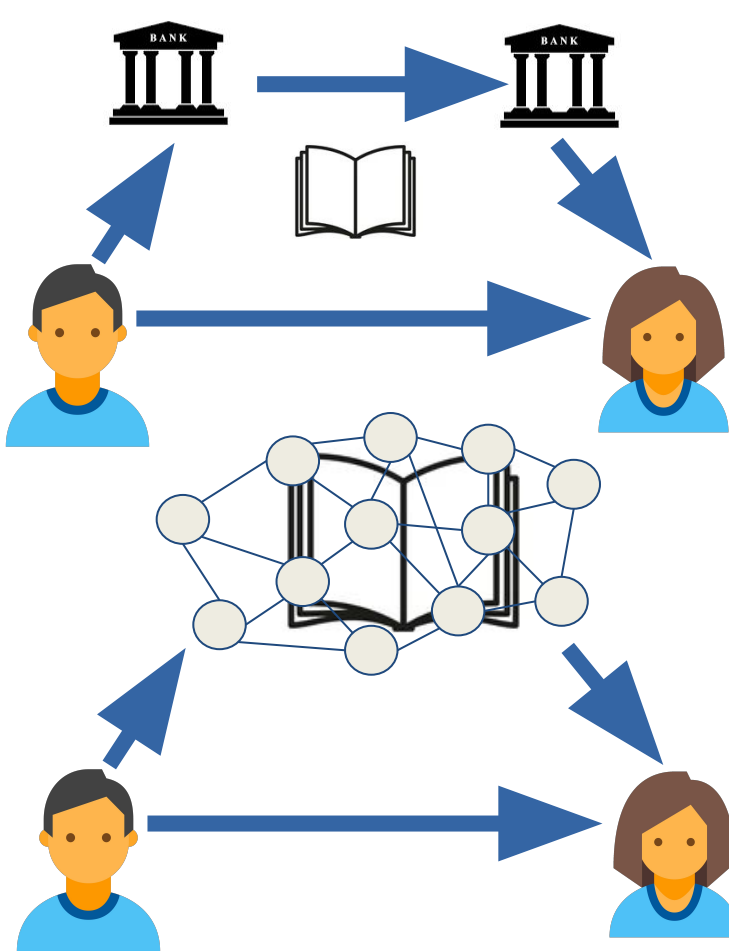
EASY

What transactions are recorded and in which order (CONSENSUS)?

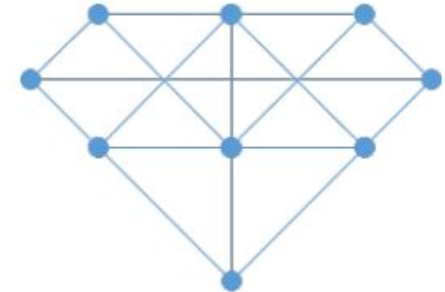
- ☐ permissioned
- ☐ permissionless

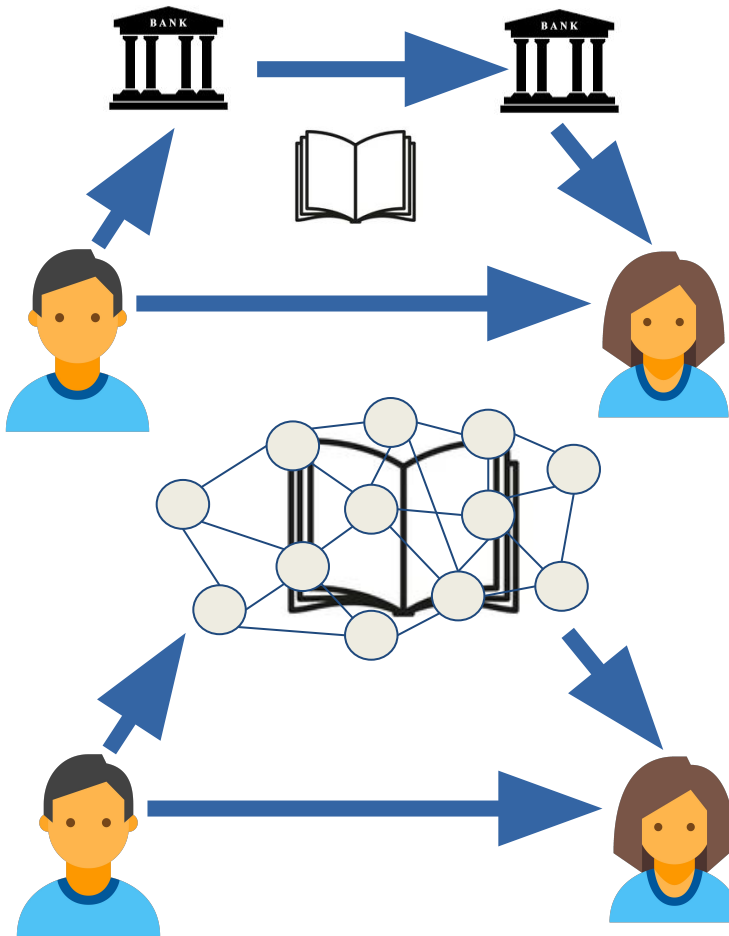
Who can read the transaction written in chain or make new proposals?

- ☐ private
- ☐ public



VS

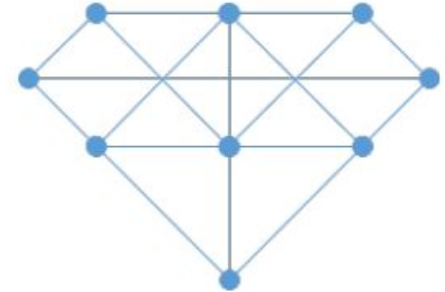




(Individuals)
TRUST

VS

CONSENSUS
(Society)

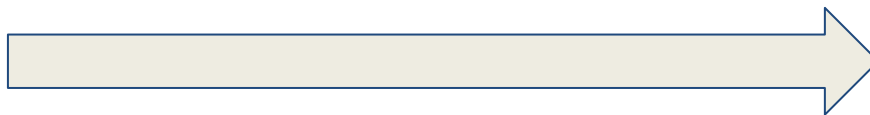


SIMPLE EXPERIMENT: what do you observe?

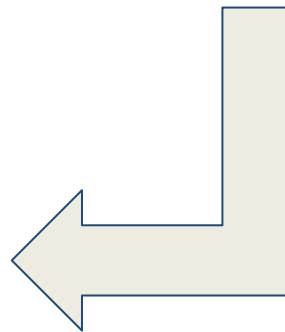
- Yahoo.co.jp
 - nslookup Yahoo.co.jp → 183.79.135.206
 - curl http://api.geoiplookup.net/?query=183.79.135.206
 - ping 183.79.135.206 -c 20
-
- Do the same with www.uniroma1.it

FLIGHT	DESTINATION	GATE #	STATUS
AA1	LOS ANGELES	A23	DELAYED
AA2	PHOENIX	C78	DELAYED
AA3	PHOENIX	B34	DELAYED
AA4	PHOENIX	A14	DELAYED
AA5	PHOENIX	C89	DELAYED
AA6	PHOENIX	G12	DELAYED
AA7	PHOENIX	C5	DELAYED
AA8	PHOENIX	D13	DELAYED
AA9	PHOENIX	A4	DELAYED
AA10	PHOENIX	B22	DELAYED
AA11	PHOENIX	B22	DELAYED

**DELAY
or MISS
or ...**



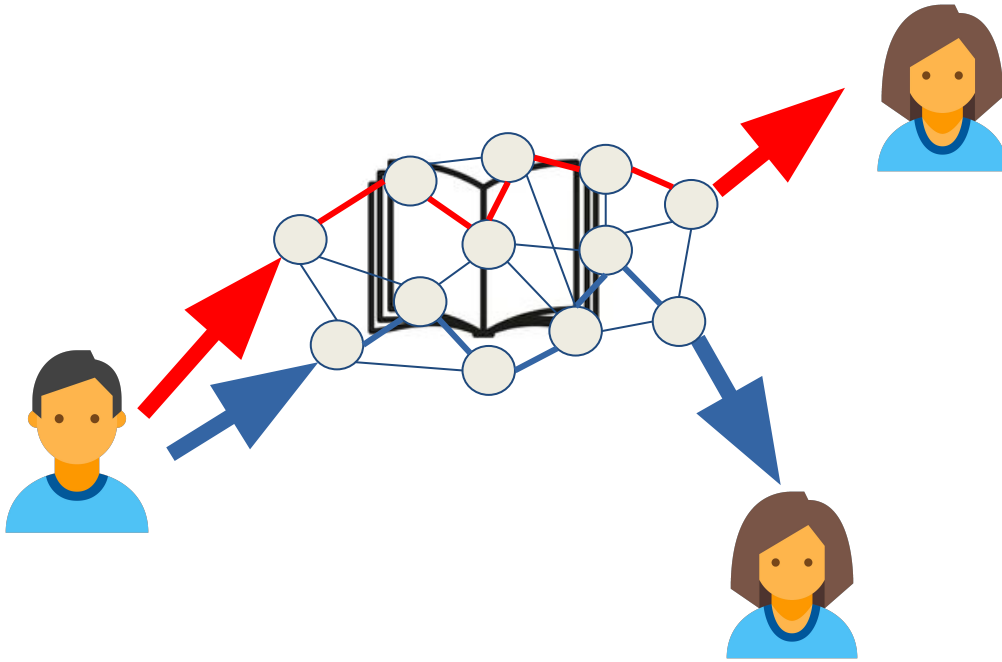
**INFORMATION
ASYMMETRY**



CONSENSUS



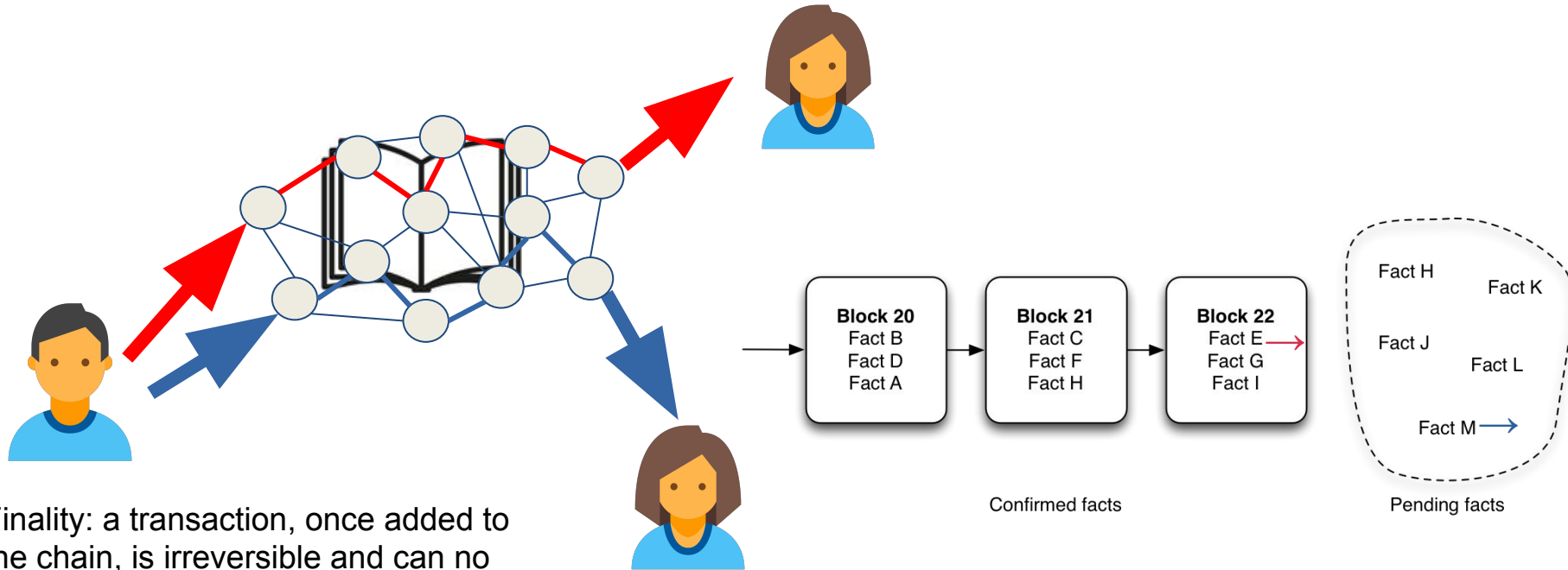
Order of events



two inconciliable events ($\rightarrow$ vs $\rightarrow$)

Order the events

If two inconciliable events (→ vs →) happens
only the first “recorded” (→) is considered legitimate



Finality: a transaction, once added to the chain, is irreversible and can no longer be altered or reversed
: deterministic vs probabilistic

Transaction

WHEN

WHAT

Block

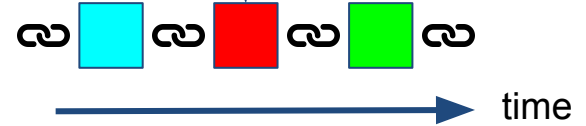
1 BTC

FROM

TO

Andrea

Maria





“If the majority of the money is in honest hands, the system works,” Micali says.

But how can he guarantee that the majority of users will act benevolently?

“If a society doesn’t have an honest majority, it no longer exists,” Micali replies. “In our society, do we have criminals? Absolutely,” but it’s a relatively small percentage, he adds. “If nobody follows our rules, there is no society.”

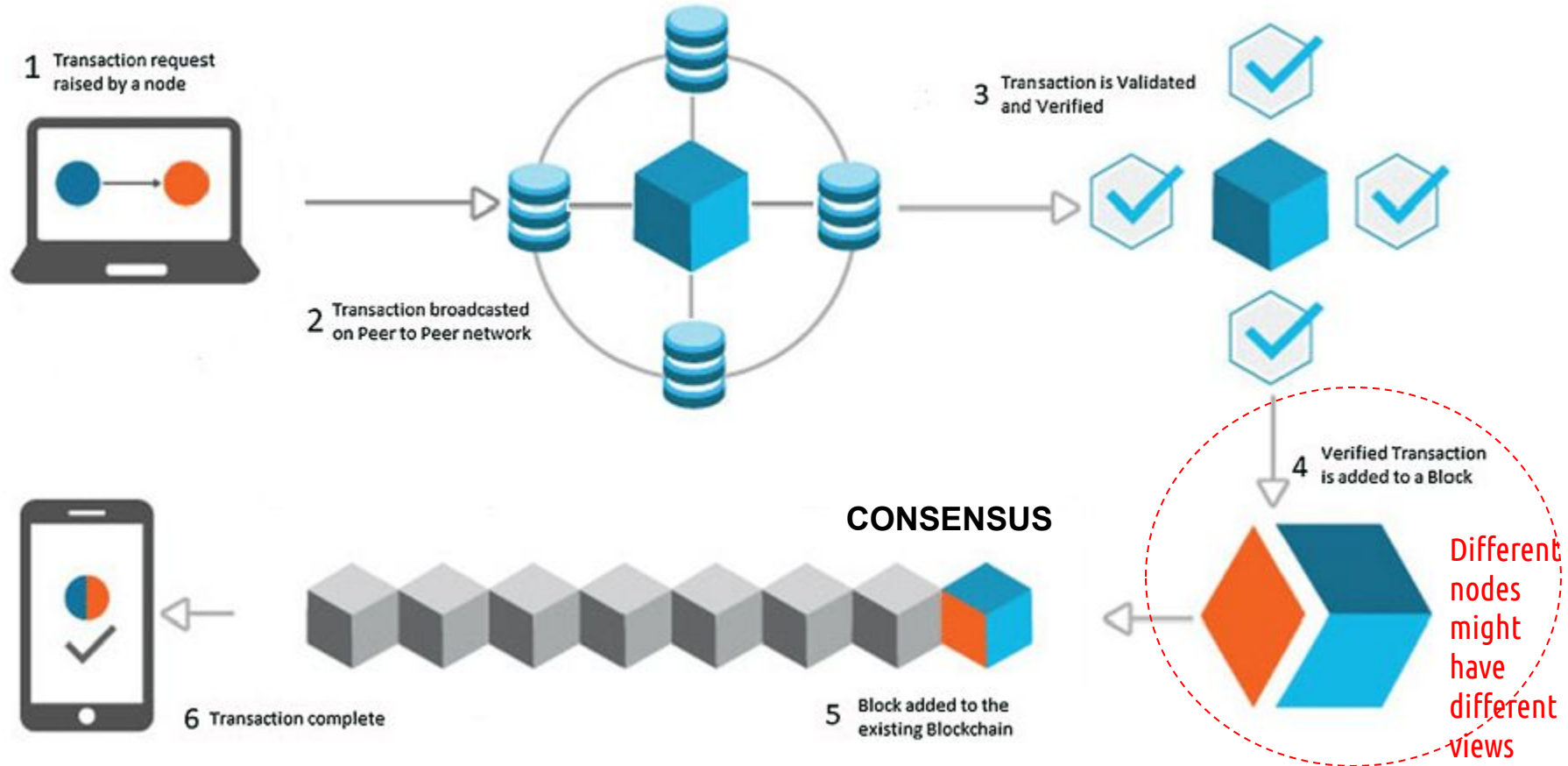
How to select what blocks are recorded in the blockchain and what is their order?

The problem of consensus

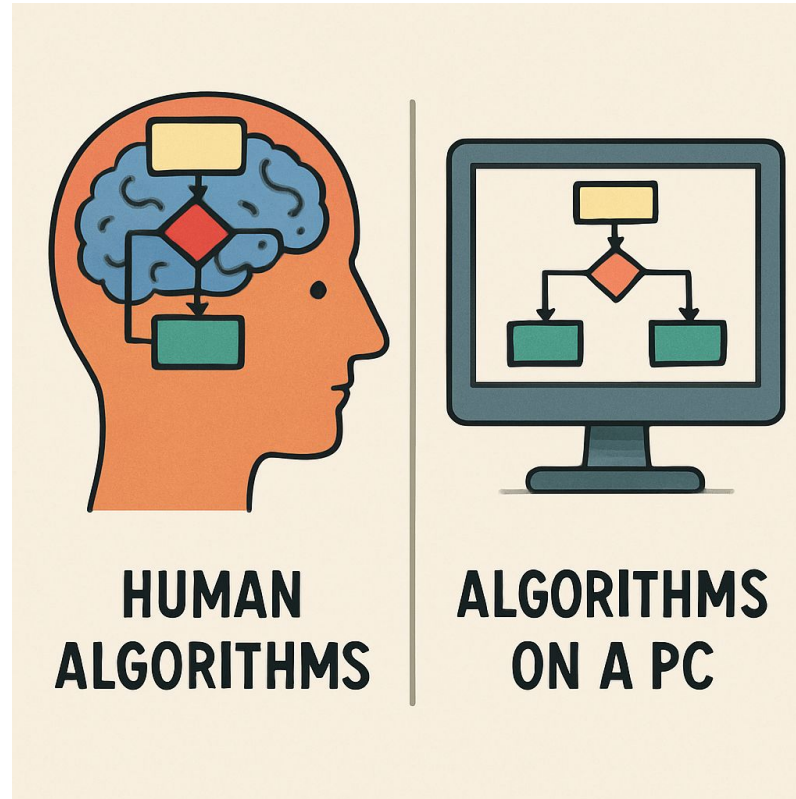
- A Blockchain Validator performs validation by verifying that **transactions are legal** (e.g. you have enough coins). ← **it simply checks the information is consistent with the protocol (layer 1), not whether it is genuine with respect to other contexts**
- Consensus involves determining the ordering of events in the blockchain — and coming to agreement on that order.

*Essentially, Consensus involves agreeing on the **ordering** of validated transactions → **fair selection** of blocks ← possible attack*

How does a transaction get into the blockchain

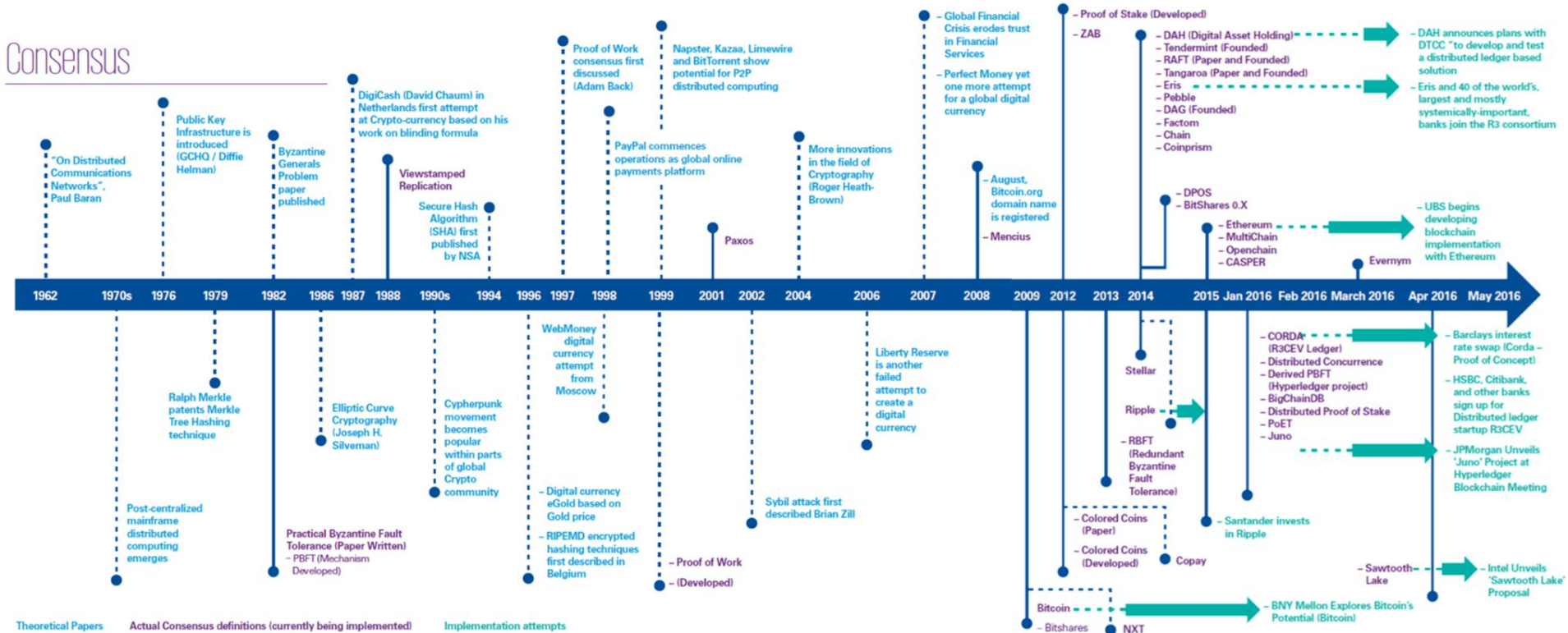


How to achieve consensus

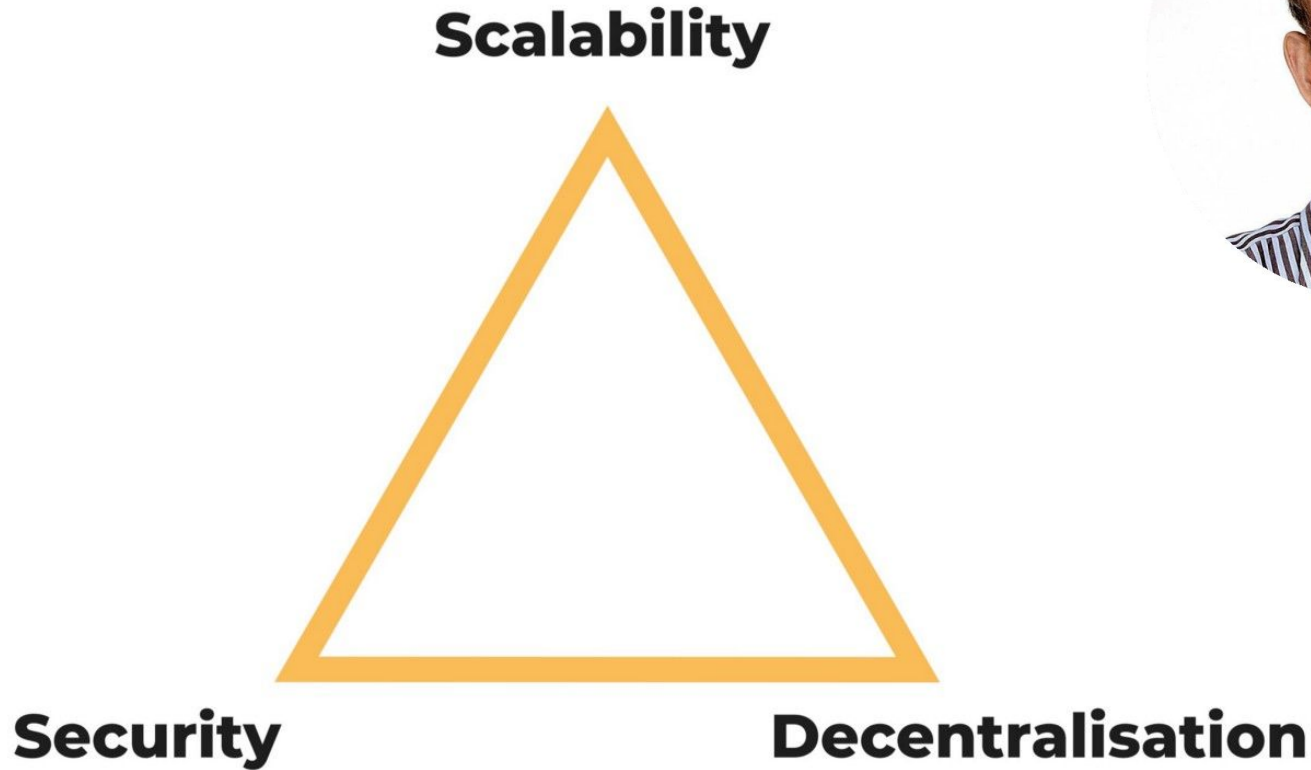


Algorithms **vs** (for) Humans

Consensus



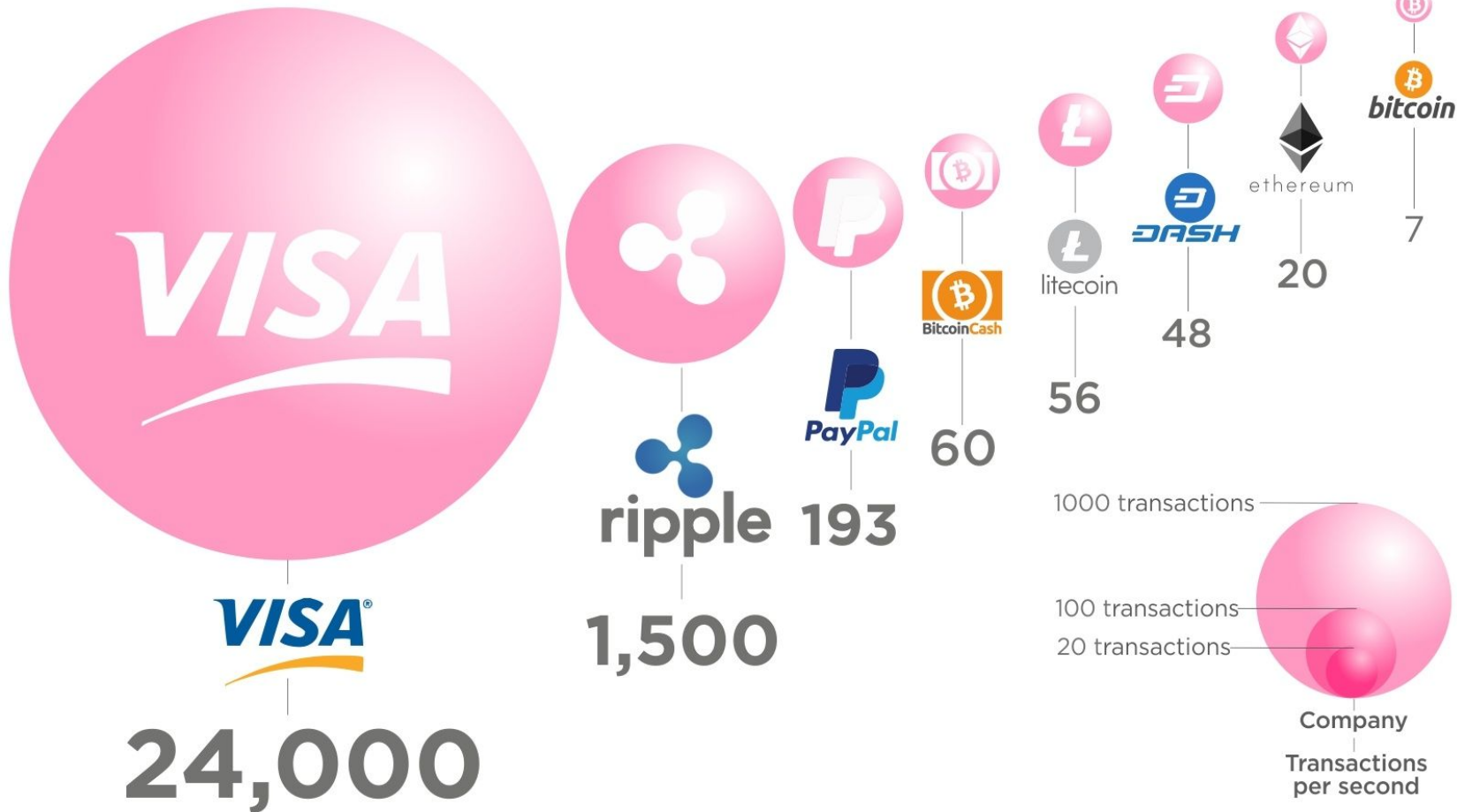
Blockchain Trilemma



Cryptocurrencies Transaction Speeds Compared to Visa & Paypal



SAPIENZA
UNIVERSITÀ DI ROMA



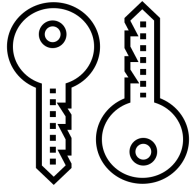
SOURCE <https://cdn.howmuch.net/articles/crypto-transactions-compared-ce3b.jpg>

Technologies of a Blockchain

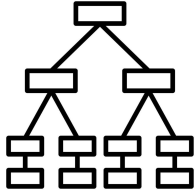


101001011

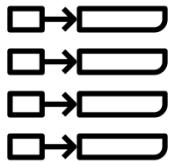
Hash Functions



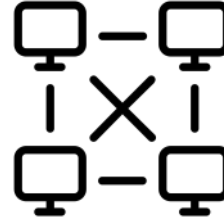
Asymmetric Encryption



Merkle Trees



Key-value Databases

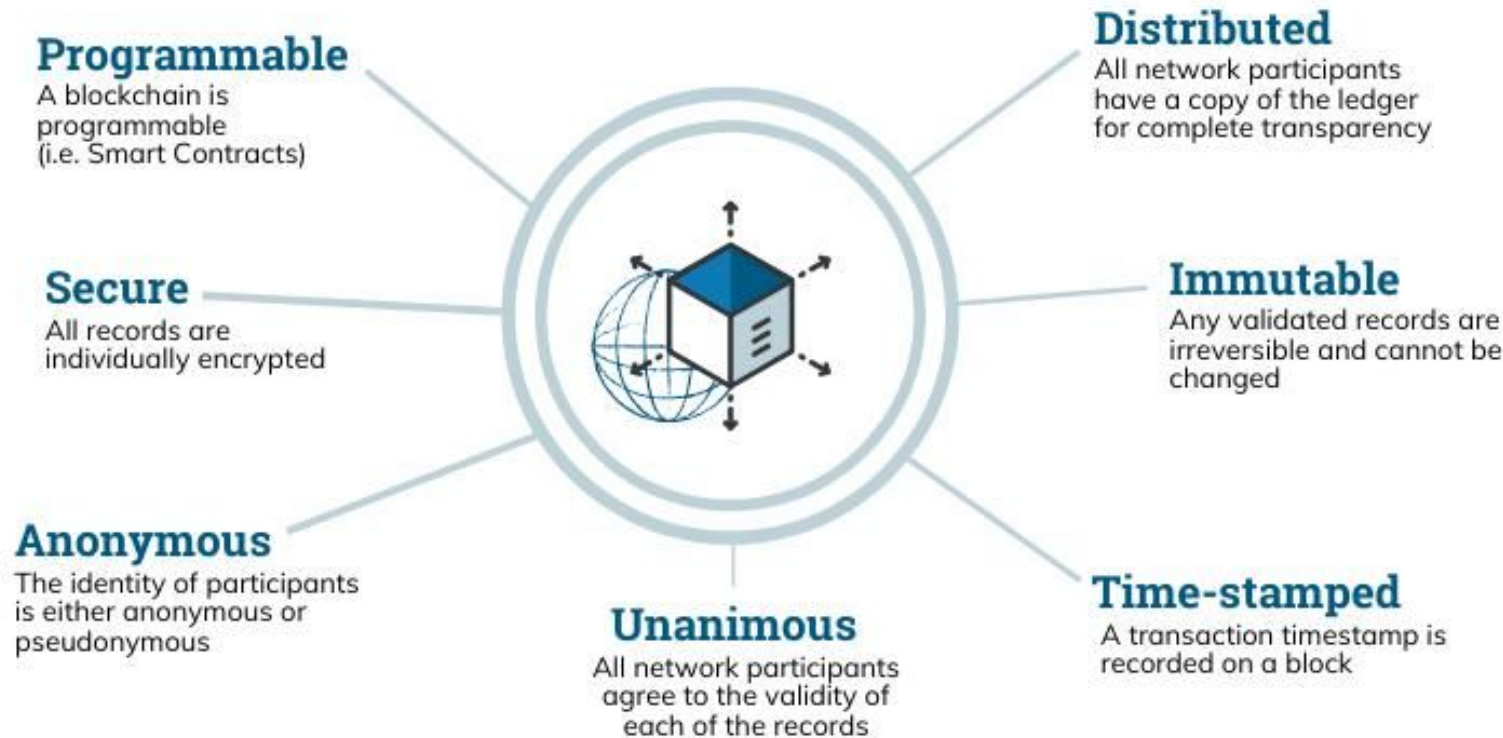


P2P communication
protocols



Consensus

The Properties of Distributed Ledger Technology (DLT)



Censorship Resistant



<https://news.bitcoin.com/deploying-censorship-resistance-to-uphold-decentralization/>

CRUD → CRAB

Burn: Deleting something from a blockchain conflicts with the immutability. We can stop the ability to `TRANSFER` the asset by transferring it to an unspendable public key. We generated an artificial public key that looks like this: `BurnBurnBurnBurnBurnBurnBurnBurnBurnBurnBurnBurnBurnBurnBurn`. The likelihood to generate a vanity address (and know the private key) that is 11 times “Burn” is extremely low.

It is very clear that the **data itself is never deleted**. Only the keys to **control the transfer of data are lost** in a burn operation.



Blockchain and the General Data Protection Regulation

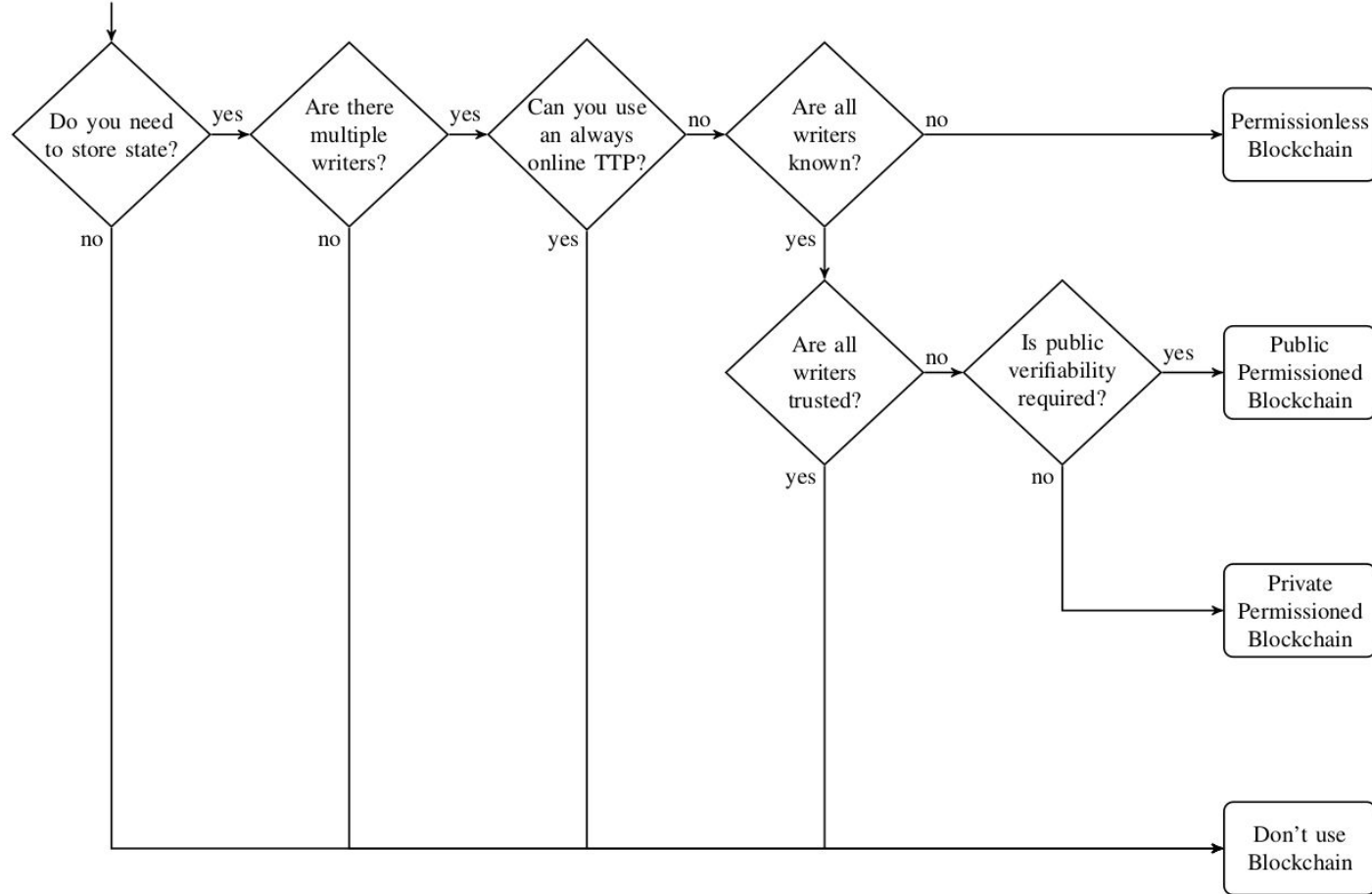
Can distributed ledgers
be squared with
European data
protection law?

A rich, but in many case unmet narrative



Do We need Blockchain?

<https://eprint.iacr.org/2017/375.pdf>



Current situation ... at least in Italy



Internet of Value: cryptocurrencies, stablecoins and Central Bank Digital Currency (CBDC);



Blockchain for business: impact of **Blockchain** technologies and **tokenization** on business ecosystems and processes

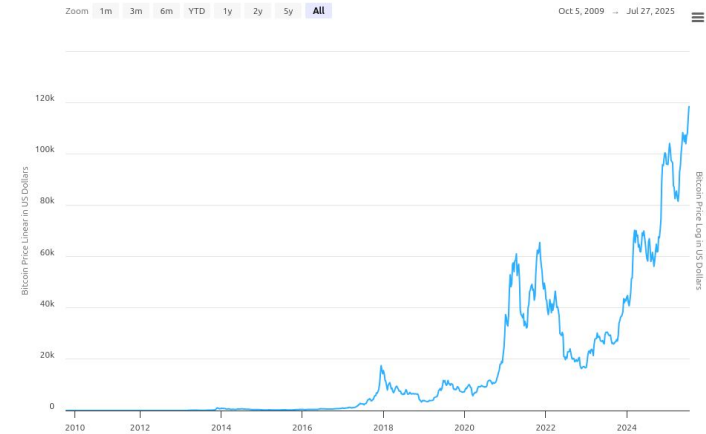


Decentralized web



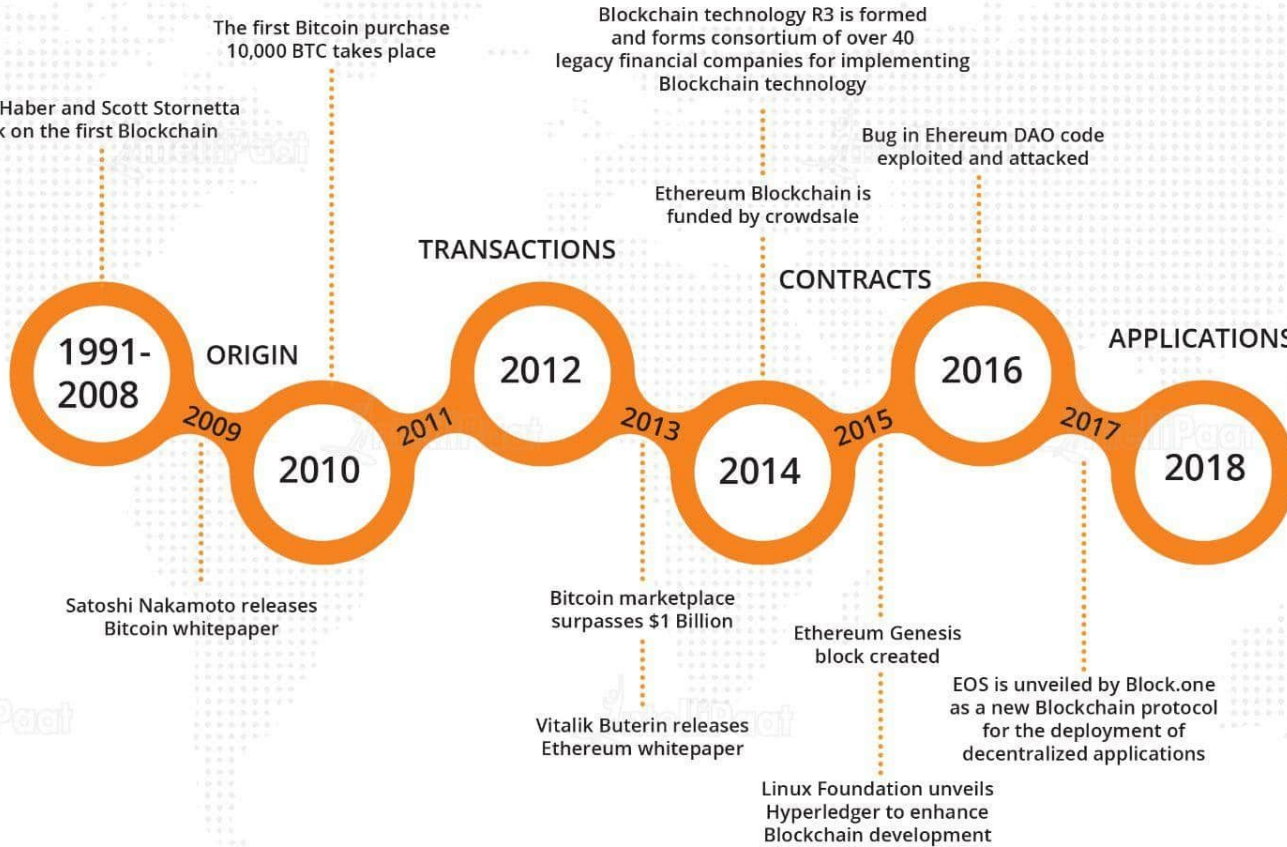
<https://www.osservatori.net/blockchain-web3/>

Bitcoin Price History Chart (Since 2009)



<https://charts.bitbo.io/price/>

THE HISTORY OF BLOCKCHAIN TECHNOLOGY



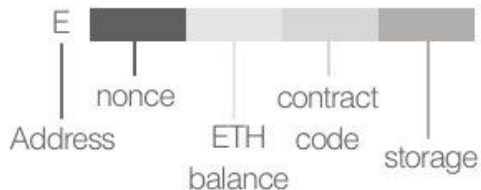
Ethereum State Transition System



State

A				
B				
C				
D				
E				

Ethereum Account



State Transition Function

Transaction

Txn Data		Signature
To:	D	
From:	B	31cb45a33ed0975ef13c
ETH:	2	
Data:	[object]	

State'

A				
B				
C				
D				
E				

State - current status of all Ethereum accounts. An account can be controlled by private keys or contract code

Transaction - contains core cryptocurrency functionality plus an optional data field, accessible by smart contracts

State Transition Function - applies a transaction the current state, progressing the EVM; capable of processing arbitrary code

```
pragma solidity >=0.7.0 <0.9.0;

/**
 * @title Storage
 * @dev Store & retrieve value in a variable
 */
contract Storage {

    uint256 number;

    /**
     * @dev Store value in variable
     * @param num value to store
     */
    function store(uint256 num) public {
        number = num;
    }

    /**
     * @dev Return value
     * @return value of 'number'
     */
    function retrieve() public view returns (uint256){
        return number;
    }
}
```

<https://etherscan.io/contractsVerified>

- ❑ Computer programs that may encode agreements, policies, rules and penalties that can not be arbitrarily altered once agreed and autonomously run on the blockchain.
- ❑ Transfer digital assets between parties

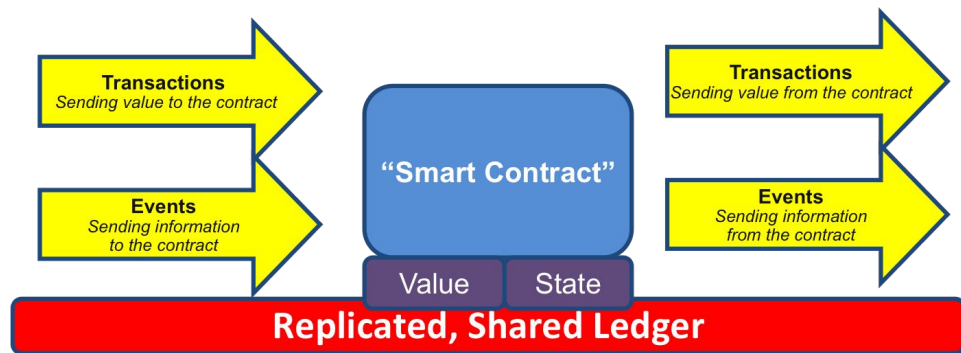
<https://remix.ethereum.org/>



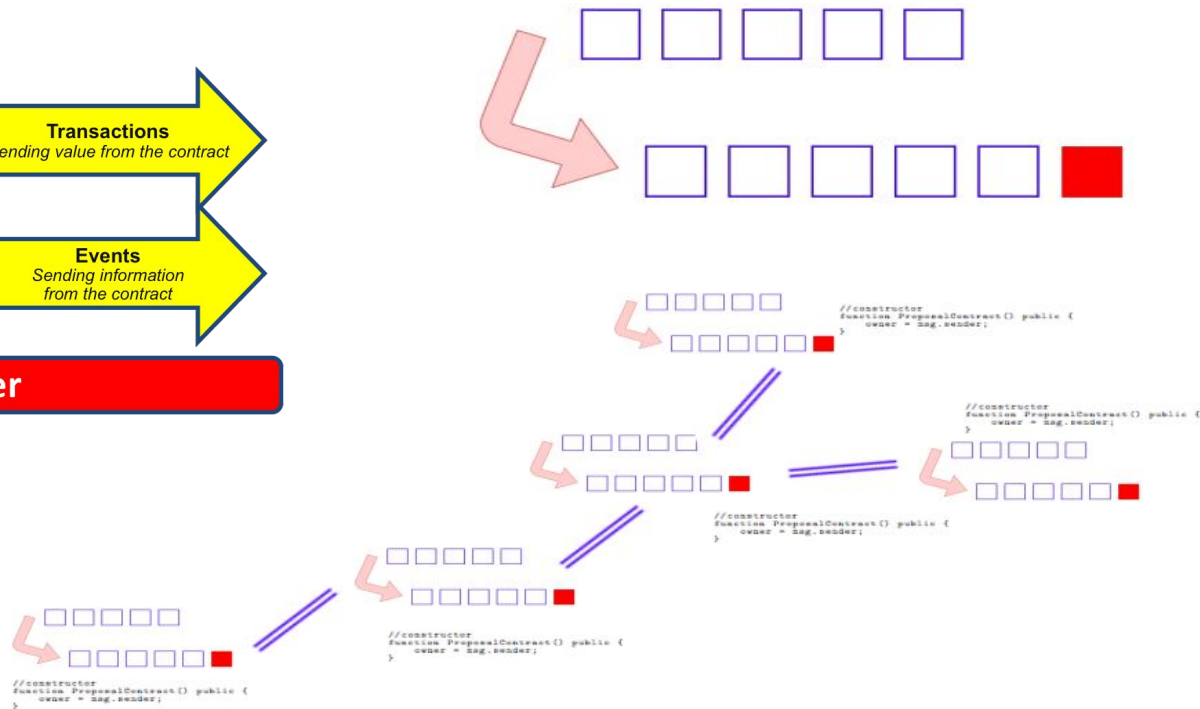
<https://medium.com/remix-ide/simple-contract-with-remix-7285af713d99>

<https://docs.soliditylang.org/en/v0.5.3/solidity-by-example.html>

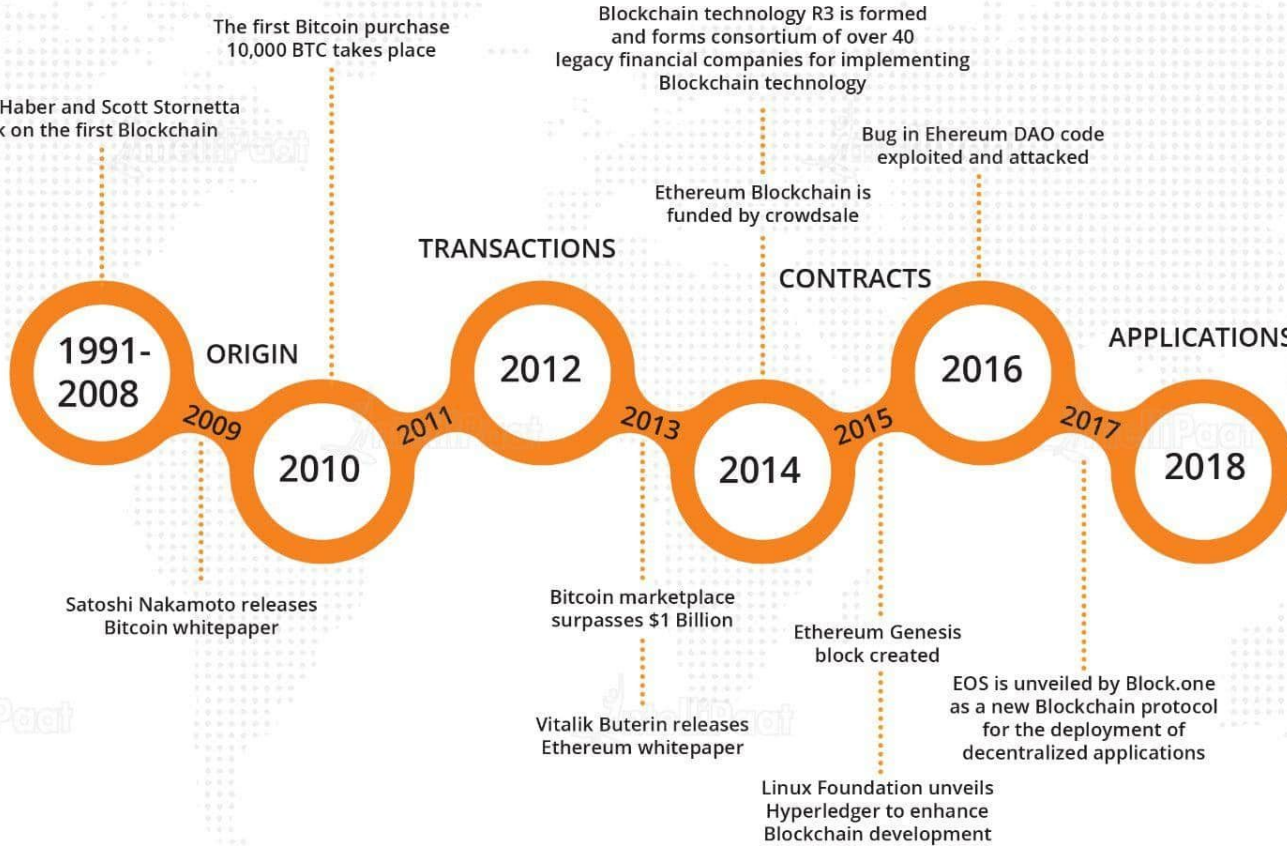
The "world computer" a P2P consensus-based state machine



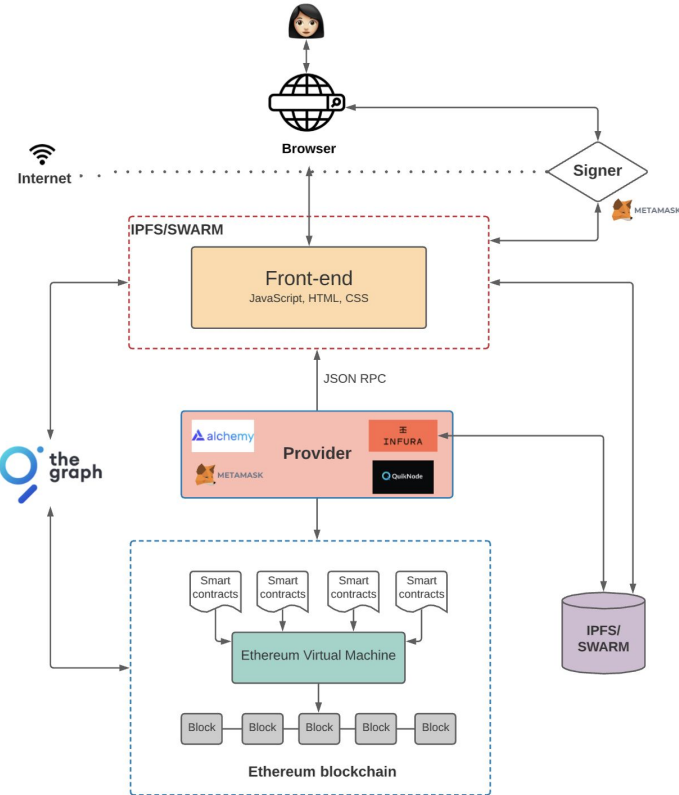
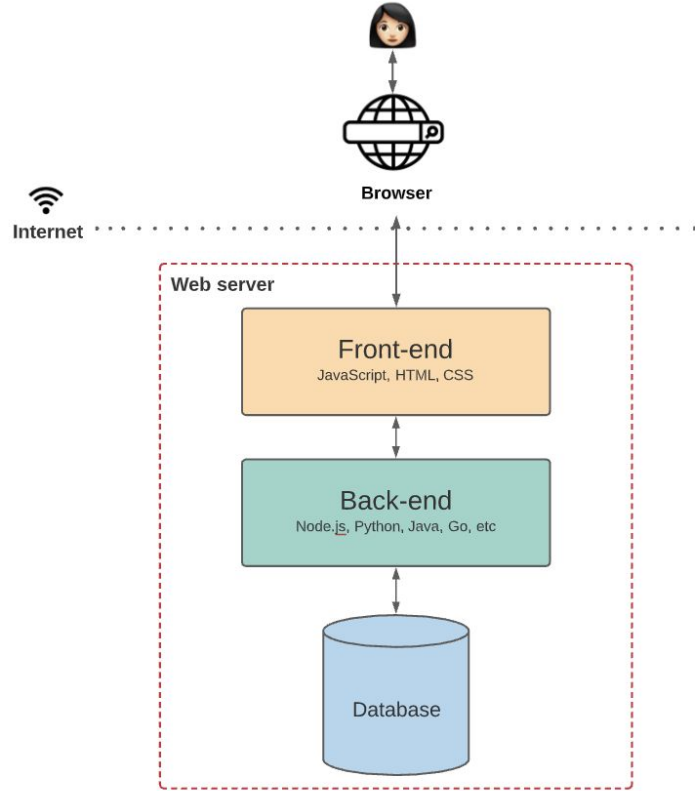
Source: gendal.me



THE HISTORY OF BLOCKCHAIN TECHNOLOGY



From App to Dapp the "world computer": Web3



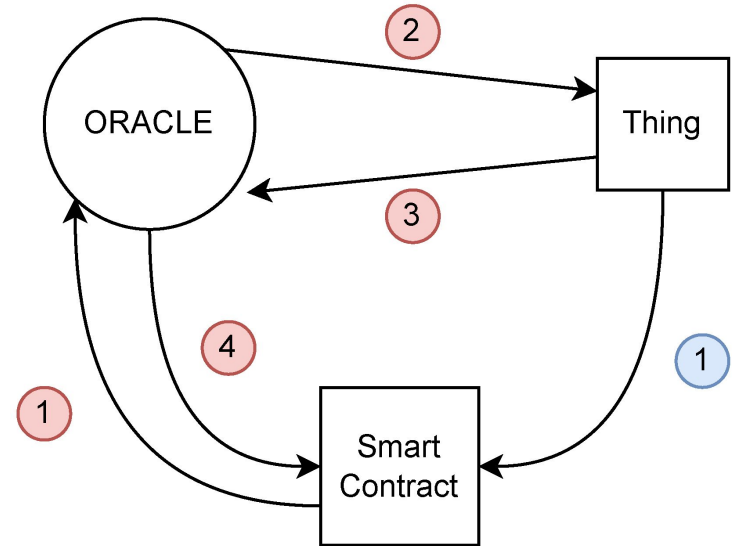
<https://www.preethikasireddy.com/post/the-architecture-of-a-web-3-0-application>

The need for a specific component: oracles

Blockchain technologies are designed to be “deterministic” to achieve consensus

If a smart contract requires accessing the measure of a smart meter, the value could differ from time to time...

Oracles are components that allow a blockchain, or a smart contract, to get inputs from outside the blockchain. They inject data coming from outside the blockchain into regular blockchain transactions. In this way, they become part of the blockchain history and can be handled deterministically by all blockchain nodes.



Can I prove what is in the chain is genuine (it is true)

very simply no you cannot!

but ... remember when you cheat in the blockchain is there forever ... your reputation is at risk

“I have no problem in dealing with assets that are created by trusted entities (i.e. centralised) provided that their management is fully decentralised”



self sovereign identity is a good example

“Blockchain (public/permissionless) is undoubtedly superior when transparency is a requirement”





<https://www.youtubeeducation.com/watch?v=HzxKs-Jd0H4>